

# Adversarial Attacks and Defenses on Graph-aware Large Language Models (LLMs)

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## Abstract

Large Language Models (LLMs) are increasingly integrated with graph-structured data for tasks like node classification, a domain traditionally dominated by Graph Neural Networks (GNNs). While this integration leverages rich relational information to improve task performance, their robustness against adversarial attacks remains unexplored. We take the first step to quantify the vulnerabilities of graph-aware LLMs by measuring the accuracy drop induced by existing adversarial attack methods tailored for graph-based models, including those for poisoning (training-time attacks) and evasion (test-time attacks), on two representative models, LLAGA (Chen et al. 2024) and GRAPH PROMPTER (Liu et al. 2024). Across the Cora, Citeseer, and PubMed benchmarks, evasion attacks are markedly more damaging than poisoning: structural perturbations reduce accuracy by as much as 51%, whereas poisoning stays within 11%. We further discover a new attack surface for LLAGA in which injecting malicious nodes as placeholders into the node sequence template drives accuracy down to as low as 0.24 under evasion, outperforming standard structural attacks. Our systematic, head-to-head comparison quantifies how specific design choices in graph encoding widen the attack margin, with three consistent findings: (1) the node sequence template in LLAGA increases its vulnerability, yielding the largest degradations of the two models; (2) the GNN encoder used in GRAPH PROMPTER is consistently more robust to structural perturbations; and (3) both approaches remain acutely susceptible to imperceptible feature perturbations, which cut accuracy by up to 73% and, when combined with structural attacks in a unified setting, by up to 87%. Finally, we propose an end-to-end defense framework GALGUARD, that combines an LLM-based feature correction module to mitigate feature-level perturbations with adapted GNN defenses against structural attacks; it recovers evasion accuracy on Cora from 0.44 to 0.83 for LLAGA, consistently across all datasets and attack types<sup>1</sup>

## 1 Introduction

Graphs, characterized by nodes and edges that represent entities and relationships, are used to model complex structures in various real-world domains, including social networks (Fan et al. 2019), biology (Dong, Mücke, and Khosla 2022), finance (Wang et al. 2021), and health-care (Ahmed-Aristizabal et al. 2021). Graph Neural Networks (GNNs) (Hamilton, Ying, and Leskovec 2017; Kipf

and Welling 2022; Veličković et al. 2018), designed specifically for graph-structured data, excel in tasks such as node classification and link prediction. Recently, the success of large language models (LLMs) such as GPT-4 (Achiam et al. 2023), Gemini (Team et al. 2023), LLaMA (Touvron et al. 2023; Dubey et al. 2024), Vicuna (Chiang et al. 2023), and PaLM-2 (Anil et al. 2023), trained on vast amounts of textual data, has sparked interest in their potential to enhance graph-related tasks. This fusion of graph data with LLMs (called *graph-aware LLMs* (Xie et al. 2023)) has opened a new line of research, where LLMs are increasingly being utilized to perform tasks traditionally handled by GNNs (Ye et al. 2023; Fatemi, Halcrow, and Perozzi 2023; Kuang et al. 2023; Zhang et al. 2023; Chen et al. 2023b, 2024). In fact, recent research has shown that graph-aware LLMs can measurably outperform GNNs on various graph-related tasks (Wenkel, Wolf, and Knyazev 2023; Chen et al. 2023b; Zhu et al. 2024; Chen et al. 2024), positioning them as powerful alternatives. Yet, as we show quantitatively, this same accuracy advantage comes paired with an unmeasured robustness cost.

Despite their growing popularity, the vulnerabilities of graph-aware LLMs to attacks remain largely unexplored, and, crucially, unquantified. Existing adversarial attacks on GNNs, including those for poisoning (training-time attacks) and evasion (test-time attacks) (Zügner, Akbarnejad, and Günnemann 2018; Zügner and Günnemann 2019; Dai et al. 2023), operate under a black-box setting where the attacker has no knowledge of the model’s internal workings. These attacks undermine *model integrity* by measurably degrading classification accuracy through subtle alterations to graph structure or node features. While GNN research has predominantly focused on structural attacks, the textual nature of graph-aware LLMs makes imperceptible perturbations to node features a significant, and as our measurements confirm, the single most damaging concern, one that has been largely overlooked.

Motivated by these three key observations: the shift from GNNs to LLMs for graph tasks, the black-box nature of current adversarial attacks on GNNs, and the potentially new vulnerabilities to feature perturbations, we ask the critical question of whether *adversarial attacks designed for GNNs remain effective against the emerging graph-aware LLM paradigm*. In addition, the success of graph-aware LLMs in handling graph tasks *highly depends* on the effective inte-

<sup>1</sup>Our code will be released upon acceptance.

gration of graphs into the LLM. While processing the node features is straightforward given LLM’s proficiency in handling textual data, incorporating the graph structure is more complex and requires innovative techniques to capture relational information (Chen et al. 2024; Guo et al. 2023; Liu and Wu 2023; Zhang et al. 2023; Sun et al. 2023; Bi et al. 2024; Pan et al. 2024). Therefore, to answer our research question with controlled, comparable measurements, we first categorize existing graph-aware methods into two classes based on their approach to encoding graphs into LLMs; one that represents graph data as *textual descriptions*, converting graph information into natural language, and another that utilizes *learned projectors* to bridge the inherent differences between graph and text domains. While conventional text-based adversarial attacks on LLMs should generally suffice to compromise LLMs handling textual descriptions of graphs, we focus our investigation on learned projector methods, using two computationally efficient representative models that differ precisely in how they encode structure, LLAGA (Chen et al. 2024) and GRAPH PROMPTER (Liu et al. 2024), both of which use a frozen pre-trained LLM. This paired design lets us attribute the measured differences in robustness directly to the encoding choice.

Our analysis quantifies how certain design choices in graph encoding for LLMs can unexpectedly enlarge the success margin of attacks. Notably, we identify a new attack surface in LLAGA that allows an attacker to drive accuracy down to as low as 0.24 by injecting malicious nodes as placeholders within the node sequence template, a larger drop than any standard structural attack achieves on the same model. In addition, we propose an end-to-end graph-aware LLM defense, GALGUARD, which integrates an LLM-based feature corrector with adapted GNN-based structural defenses and consistently restores accuracy across all datasets and attack types. We are the first to investigate, and to quantify, the vulnerability of graph-aware LLMs to adversarial attacks. The contributions of the paper are summarized as follows:

- We leverage representative adversarial attacks for GNNs, specifically NETTACK (Zügner, Akbarnejad, and Günnemann 2018) and METAATTACK (Zügner and Günnemann 2019), to investigate the transferability of poisoning and evasion attacks to graph-aware LLMs.
- We create a taxonomy for graph-aware LLMs based on their encoding approaches to understand how different methodologies integrate graph structures into LLMs.
- We discover a new attack surface in LLAGA, showing that injecting malicious nodes as placeholders into the node sequence template can degrade its performance.
- We provide a holistic assessment of the vulnerabilities of graph-aware LLMs by performing imperceptible attacks on textual node features, a perspective often overlooked in GNNs.
- We propose GALGUARD, a novel end-to-end defense strategy that integrates LLM-based feature correction with the adaptation of existing GNN-based structural defenses.

## 2 Graph Encoding-Based Taxonomy of LLM Adaptations

Graph encoding methods for LLMs can be categorized into two main approaches: *textual descriptions of graphs* (where graph information is encoded as natural language) and *learned graph projectors* (where a projector is utilized to encode graph data into embeddings or structured forms that LLMs can directly process). We briefly describe the different encoding approaches in each category. Figure 5 provides the illustration of both approaches. Full description in Appendix A.1 and related work on GNN attacks in Appendix A.2.

**Graph Representation as a Textual Description.** This approach converts graph data into natural language, enabling LLMs to process them as textual descriptions, often with specific instructions to query LLMs (Guo et al. 2023; Liu and Wu 2023; Ye et al. 2023; Wenkel, Wolf, and Knyazev 2023; Wei et al. 2024b; Fatemi, Halcrow, and Perozzi 2023; Tan et al. 2023; Zhao et al. 2023; Wei et al. 2024a; Zhang 2023). In multimodal applications, Liu et al. (2023) and Su et al. (2022) linked molecule graphs with textual descriptions. Furthermore, Huang et al. (2023a) investigated incorporating graph structure into LLM prompts. Their findings reveal that LLMs tend to process graph-related prompts as contextual paragraphs, underscoring the importance of prompt design, as seen in studies like (Brannon et al. 2023; Zhang et al. 2023; Sun et al. 2023; Bi et al. 2024; Pan et al. 2024; He et al. 2024). Other research has focused on using LLMs to enhance node features, graph structures, act as predictors, or serve as feature extractors for GNNs (Chen et al. 2023a; Chandra et al. 2020; Chen et al. 2023b; Mavromatis et al. 2023). For inference, Zhu et al. (2024) proposed a fine-tuning method for textual graph that combined LLMs and GNNs, and Duan et al. (2023) introduced an effective approach that enhances textual graph learning. While these methods offer a straightforward way to use LLMs for graph tasks, they often struggle to fully capture structural intricacies (especially for large graphs) and their reported gains depend heavily on prompt engineering.

**Graph Representation as a Learned Projector.** To address the limitations of textual representations, learned projector methods transform graph data into embeddings or graph vectors that LLMs can process more directly. This includes methods like LLAGA, which uses node-level templates and a linear projector (Chen et al. 2024), and others that incorporate GNN layers into a pre-trained LLM (Liu et al. 2024; Qin et al. 2023). Other approaches use graph neural prompts to encode knowledge graphs (Tian et al. 2024), develop prompt-based node feature extractors with a GNN adapter (Huang et al. 2023b), or condense graph information into fixed-length prefixes using graph transformers (Chai et al. 2023; Yang et al. 2021; Peng et al. 2024).

**This Work:** We focus on projector methods that utilize *frozen pre-trained LLMs* because they are computationally efficient and avoid the substantial costs of retraining (Chen et al. 2024; Liu et al. 2024; Qin et al. 2023; Tian et al. 2024; Huang et al. 2023b). For a clean, controlled comparison, we select two representative models that share the frozen-LLM design but differ in exactly one axis, the structure encoder:

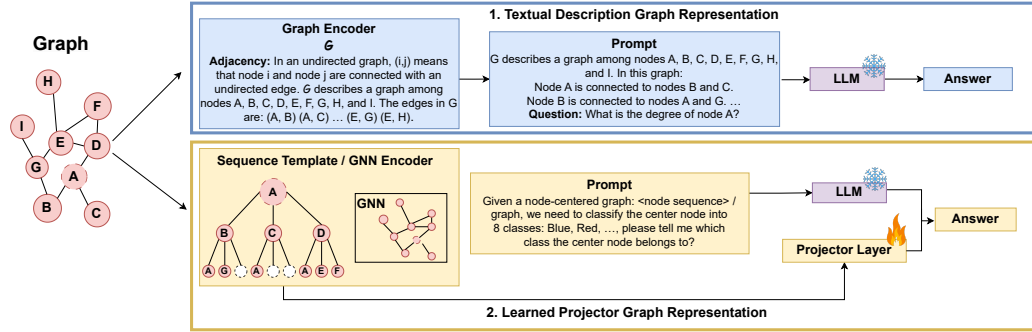


Figure 1: **Graph encoding-based adaptations of LLMs.** In the textual description graph representation, graphs are explained textually (top part). In the learned projector representation, graphs are encoded either by node templates that turns the graph into sequences or by GNNs (bottom part). Note that in both cases, the LLM is frozen. Only the projector that maps the graph into tokens is trained.

LLAGA (Chen et al. 2024), which employs node-level templates with a linear projector, and GRAPHPROMPTER (Liu et al. 2024), which uses a GNN with a linear projector. Isolating this single variable is what lets our experiments attribute the resulting gaps in attack success to the encoding choice rather than to confounding factors. We do not consider textual description methods, as they often struggle with scalability and do not introduce the novel, graph-specific vulnerabilities we aim to investigate.

### 3 Threat Model

We characterize our threat model based on established research on adversarial attacks in graphs, and extend its applicability to graph-aware LLMs (Zügner, Akbarnejad, and Günnemann 2018; Wu et al. 2019; Zügner and Günnemann 2019; Ma, Ding, and Mei 2020; Jin et al. 2021; Dai et al. 2023; Li et al. 2023). For example, in social networks, attackers often control only a few bot accounts; this limited access aligns with our threat model, where adversaries manipulate a small subset of nodes to evade detection or influence legitimate user classification. Another example is Wikipedia, where hoax articles with sparse connections to legitimate ones can be strategically linked to manipulate the classification of genuine content, causing misclassification of real articles’ categories. This transferability is particularly concerning as these advanced models have the potential of being deployed in critical applications such as recommendation system, social media analysis or even healthcare, potentially replacing GNNs. Thus, following existing works on adversarial attacks in graphs, we characterize the threat model along three dimensions; model access, adversarial capabilities, and attack strategies, followed by defining the attacker’s goal.

**Model Access.** The attacker operates in a black-box setting, with no knowledge of the model’s architecture or parameters. The attacker can only influence the model indirectly by interacting with the training or test data.

**Adversarial Capabilities.** Consistent with existing poisoning (evasion) attacks, the attacker has complete access to the training (inference) data of the target model. Within a predefined budget and under unnoticeability constraints (aiming to

avoid detection), the attacker can modify or inject data.

**Attack Strategies.** The attacker leverages two primary strategies: feature manipulation (altering node features) and structure manipulation (adding or removing edges).

**Attacker’s Goal.** The attacker’s goal is to compromise model integrity by measurably degrading overall performance, specifically by increasing the misclassification rate of the classification model. Throughout, we quantify this goal directly as the drop in node-classification accuracy relative to the clean model, which provides a single, comparable yardstick across attacks, models, and datasets.

### 4 Experimental Setup

**Dataset.** We evaluate on three widely used text-attributed graph benchmarks, Cora, Citeseer, and PubMed (Yang, Cohen, and Salakhudinov 2016), chosen because they span a range of scales and densities (see the sparsity column of Table 1) and are standard benchmarks for models like those in our study (Chen et al. 2023a, 2024; Liu et al. 2024). This spread lets us measure how attack success varies with graph density in a controlled way. To probe scalability, we additionally report on the far larger ArXiv dataset (Hu et al. 2020) in Appendix A.5. We adopt standard train/validation/test splits of 6:2:2 for the three smaller datasets and 5:2:3 for ArXiv, as in (Chen et al. 2024). Detailed statistics are provided in Table 1.

Table 1: Dataset statistics. Sparsity (edges per node, in %) decreases from Cora to PubMed, the same ordering along which we later observe attack success to decline.

| Data     | #Node | #Edge | Sparsity (%) |
|----------|-------|-------|--------------|
| Cora     | 2708  | 5429  | 14.81        |
| CiteSeer | 3327  | 4614  | 8.34         |
| PubMed   | 19717 | 44338 | 2.28         |

**Tasks.** We use *node classification* as our primary task, a widely used method for evaluating graph machine learning models and the primary setting for adversarial at-

tacks (Zügner, Akbarnejad, and Günnemann 2018; Zügner and Günnemann 2019; Liu and Wu 2023).

**Evaluation Metrics.** We report node-classification *accuracy*, the proportion of correctly predicted labels, as our single headline metric, and express attack strength as the percentage-point drop in accuracy relative to the clean model so that every attack, model, and dataset is placed on a common, directly comparable scale. To ensure the reported gaps are not artifacts of run-to-run noise, each experiment is repeated three times and we report the mean with its standard deviation; the tight standard deviations reported throughout confirm that the comparative margins we highlight are stable. All experiments were conducted on a machine equipped with an NVIDIA A40 GPU with 48GB of memory.

**Choice of Graph-aware LLM and Adversarial Attack Methods.** We clarify our selection of graph-aware LLM methods and adversarial attack methods used to investigate the vulnerabilities of graph-aware LLMs.

**Graph-Aware LLM Methods.** We employ two representative graph-aware LLM methods: LLAGA (Chen et al. 2024) and GRAPHPROMPTER (Liu et al. 2024). Both methods utilize a *frozen pre-trained LLM*, making them computationally efficient and adaptable to various graph tasks and LLM architectures. LLAGA uses node-level templates with a linear projector, while GRAPHPROMPTER employs a GNN-based encoder and a linear projector. In both cases, the projector is a 2-layer multi-layer perceptron (MLP). The foundational LLMs for LLAGA and GRAPHPROMPTER are Vicuna-7B (Chiang et al. 2023) and LLAMA2-7B (Touvron et al. 2023), respectively.

**Adversarial Attack Methods.** We employ the poisoning and evasion variants of two representative adversarial attack methods, NETTACK (Zügner, Akbarnejad, and Günnemann 2018) and METAATTACK (Zügner and Günnemann 2019), for conducting our investigation. NETTACK is a targeted attack that finds minimal perturbations to misclassify specific nodes, while METAATTACK performs untargeted attacks by treating the graph structure as a hyperparameter to optimize. For NETTACK, we adopt the approach of (Li et al. 2023; Zügner and Günnemann 2019) by sequentially targeting each node and aggregating the results, effectively transforming it into an untargeted evaluation. To keep the comparison fair and unnoticeable, every method operates under an identical budget of only 10% perturbed edges, as in (Zügner, Akbarnejad, and Günnemann 2018; Zügner and Günnemann 2019), so that any difference in measured accuracy drop reflects the attack strategy rather than a larger perturbation allowance. Full details in Appendix A.2.

## 5 Results

### 5.1 Poisoning and Evasion Attacks on Graph-aware LLMs

We analyze poisoning and evasion attacks by adapting methods from NETTACK (Zügner and Günnemann 2019) and METAATTACK (Zügner and Günnemann 2019). For poisoning attacks on LLAGA, we use a perturbed graph to generate the node-level sequences and then retrain the projector. For GRAPHPROMPTER, we use the perturbed graph to re-encode

the graph using a GAT and retrain its projector. For evasion attacks, both models are trained on unperturbed data, and perturbations are introduced only during the inference phase.

**Results.** Table 2 surfaces three quantitative patterns, each consistent across all three datasets and both models. **(1) Evasion dwarfs poisoning.** Poisoning leaves graph-aware LLMs largely intact, degrading accuracy by at most 11% under either attack method, indicating that the models’ learned representations and the additional context supplied by the frozen LLM absorb training-time perturbations. Evasion, by contrast, is far more potent under the identical 10% edge budget, cutting accuracy by up to 51%, an almost fivefold larger margin, which localizes the vulnerability squarely at inference time. **(2) METAATTACK is the stronger attack.** Method-for-method, METAATTACK produces a larger accuracy drop than NETTACK for both poisoning and evasion across every dataset, consistent with its treatment of graph edges as optimizable hyperparameters that yield small but highly effective perturbations. **(3) Density predicts vulnerability.** Denser graphs degrade more than sparse ones: under METAATTACK evasion, LLAGA loses 51% of its accuracy on the dense Cora graph but only 19% on the sparser PubMed, a  $2.7\times$  difference in attack success attributable to graph density alone.

### 5.2 Which Graph-aware Paradigm is More Susceptible to Graph Adversarial Attacks and Why?

The head-to-head numbers in Table 2 give a clear verdict: LLAGA is the more fragile of the two paradigms, and the gap traces directly to its *sensitivity to structural perturbations*. The contrast is sharpest under METAATTACK evasion on Cora, where LLAGA falls to 0.44 (a 51% drop) while GRAPHPROMPTER holds at 0.52 (only a 13% drop), a nearly  $4\times$  larger relative degradation for LLAGA; the same ordering holds on Citeseer (35% vs. 22%) and PubMed (19% vs. 14%). LLAGA integrates the graph structure in two ways: by generating a fixed-length node-level sequence for each node and by computing a graph Laplacian embedding. Each position in this sequence uniquely corresponds to a relative structural position within the original graph. Because the node embeddings are derived directly from the perturbed structure, the perturbation propagates undiluted into the information the LLM processes, which is why LLAGA posts the larger drops. In contrast, GRAPHPROMPTER’s textual node embeddings remain unchanged during structural perturbations, as only the GNN component is affected. This design lets the unperturbed textual features augment the perturbed structure, and the consistently smaller drops in the GRAPHPROMPTER columns quantify the robustness this buys.

## 6 Node Sequence Template Injection Attacks

We demonstrate a new attack surface in LLAGA. Specifically, by exploiting the neighborhood detail template used to construct the node sequences. The attack is illustrated in Figure 2 and the algorithm is in Appendix A.4. To obtain the node sequence for a particular node  $u$ , the neighborhood detail template in LLAGA first constructs a *fixed-shape* computa-

Table 2: Node-classification accuracy (mean $\pm$ std over three runs) under NETTACK and METAATTACK, compared head-to-head for LLAGA and GRAPH PROMPTER on Cora, Citeseer, and PubMed. The percentage in parentheses is the accuracy drop from the Clean row (higher = more damage). Two patterns are consistent across every dataset: evasion is far more damaging than poisoning (drops up to 51% vs. up to 11%), and LLAGA degrades more than GRAPH PROMPTER under structural attack.

| Attack     | Type      | Cora                  |                       | Citeseer              |                       | PubMed                |                       |
|------------|-----------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|            |           | LLAGA                 | GRAPH PROMPTER        | LLAGA                 | GRAPH PROMPTER        | LLAGA                 | GRAPH PROMPTER        |
| Clean      | —         | 0.89 $\pm$ 0.07       | 0.60 $\pm$ 0.03       | 0.71 $\pm$ 0.04       | 0.70 $\pm$ 0.03       | 0.90 $\pm$ 0.04       | 0.90 $\pm$ 0.02       |
| NETTACK    | Poisoning | 0.87 $\pm$ 0.04 (2%)  | 0.60 $\pm$ 0.03 (0%)  | 0.64 $\pm$ 0.05 (10%) | 0.68 $\pm$ 0.02 (3%)  | 0.89 $\pm$ 0.03 (1%)  | 0.88 $\pm$ 0.02 (2%)  |
|            | Evasion   | 0.55 $\pm$ 0.09 (38%) | 0.53 $\pm$ 0.05 (12%) | 0.59 $\pm$ 0.04 (19%) | 0.61 $\pm$ 0.03 (14%) | 0.84 $\pm$ 0.05 (7%)  | 0.81 $\pm$ 0.02 (10%) |
| METAATTACK | Poisoning | 0.79 $\pm$ 0.03 (11%) | 0.58 $\pm$ 0.05 (3%)  | 0.63 $\pm$ 0.04 (12%) | 0.65 $\pm$ 0.06 (7%)  | 0.89 $\pm$ 0.08 (1%)  | 0.85 $\pm$ 0.02 (6%)  |
|            | Evasion   | 0.44 $\pm$ 0.06 (51%) | 0.52 $\pm$ 0.04 (13%) | 0.50 $\pm$ 0.03 (35%) | 0.56 $\pm$ 0.02 (22%) | 0.73 $\pm$ 0.04 (19%) | 0.77 $\pm$ 0.03 (14%) |

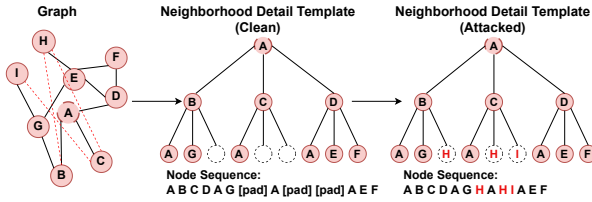


Figure 2: Illustration of the node sequence template injection attacks, the new attack surface unique to LLAGA. The example shown is the Non-adjacent Injection attack (NI); red dashed lines are the attacker’s edge perturbations. Under Supernode Injection (SI) and Multiple Supernode Injection (MSI), one or several high-degree supernodes are injected into the placeholder slots, which is the design choice that makes these strategies progressively more damaging in Table 3. Here,  $k = 3$ .

tional tree for each node, where the children of a node are its sampled 1-hop neighbors. This process is repeated to sample neighbors for each subsequent level of the tree. If a node has an insufficient number of neighbors, a placeholder is used to maintain the fixed size of e.g.  $k = 10$  children, which is consistent with LLAGA’s original design (see “Clean” from Figure 2 and lines 1-14 in the algorithm). This fixed structure provides a new attack surface.

We demonstrate a new attack surface in LLAGA by exploiting its neighborhood detail template algorithm. An attacker can craftily inject malicious nodes as placeholders through graph manipulation to significantly degrade model performance. We propose three strategies for this attack. The first, termed *Non-Adjacent Injection (NI)*, involves injecting nodes that are not within the 2-hop neighborhood of a target node  $u$  (see “Attacked” from Figure 2 and lines 21-27). The second approach, referred to as *Supernode Injection (SI)*, involves injecting a high-degree node as a placeholder, ensuring it is outside the target node’s 2-hop neighborhood. The third, *Multiple Supernode Injection (MSI)* is a variant of *SI* that injects multiple distinct supernodes as placeholders (lines 35-43). The rationale is that the inclusion of uninformative nodes in the node sequence corrupts the representation of the target node, leading to a poorly trained projection layer.

This compromises the integrity of the model’s learning and inference process.

**Results.** Table 3 reveals a stark split between attack settings. Under poisoning, LLAGA is highly resilient on Cora and Citeseer, staying at or above 0.82 and 0.62 respectively. Under evasion, however, the same three strategies collapse its accuracy to 0.42, 0.36, and 0.30 on Cora and to 0.36, 0.30, and 0.24 on Citeseer for *NI*, *SI*, and *MSI* respectively. The three strategies also form a clean, monotonic ordering of severity,  $NI < SI < MSI$ , on every dataset, confirming that injecting more and higher-degree supernodes progressively widens the damage.

The dataset trend mirrors the density pattern seen earlier: on the much larger PubMed graph, LLAGA is markedly more robust, with evasion accuracy holding at 0.85, 0.82, and 0.76, only a modest slide from the 0.90 clean baseline. The interpretation is that in the smaller, tightly connected Cora and Citeseer graphs a localized injection concentrates its impact and severely disrupts local-neighborhood reasoning, whereas on PubMed the same injection is diluted across a far larger neighborhood, blunting its effect.

Most importantly, the injection attack is the strongest evasion attack we measure on LLAGA, beating the best standard structural baseline. Head-to-head, *MSI* drives evasion accuracy down to 0.30, 0.24, and 0.76 on Cora, Citeseer, and PubMed, versus 0.44, 0.50, and 0.73 for METAATTACK (Table 2), a further 0.14 and 0.26 absolute accuracy reduction on Cora and Citeseer. This margin, achieved by exploiting the template rather than the graph at large, is direct evidence that the node sequence template is itself the exploitable weakness.

## 7 Poisoning and Evasion Attack via Feature Perturbation

Existing graph adversarial attacks primarily focus on structural perturbations (Li et al. 2023; Wu et al. 2019; Dai et al. 2018; Sun et al. 2022; Zügner and Günnemann 2019), with the exception of NETTACK (Zügner, Akbarnejad, and Günnemann 2018), which is not transferable to the textual domain. Motivated by the previous result, that structural attacks dent GRAPH PROMPTER by only 13–22% and that unperturbed textual features are what kept it robust, we now target those very features with imperceptible perturbations to test whether the feature channel is the more effective point of

Table 3: Node sequence template injection attack on LLAGA (accuracy, mean $\pm$ std). *NI*, *SI*, and *MSI* denote non-adjacent, single-supernode, and multiple-supernode injection; Clean is the un-attacked baseline. Severity increases monotonically NI  $\rightarrow$  SI  $\rightarrow$  MSI in every row, and evasion is consistently far more damaging than poisoning; PubMed’s larger graph dilutes all three attacks.

| Data     | Attack    | NI              | SI              | MSI             |
|----------|-----------|-----------------|-----------------|-----------------|
| Cora     | Clean     | ————            | 0.89 $\pm$ 0.07 | ————            |
|          | Poisoning | 0.86 $\pm$ 0.03 | 0.84 $\pm$ 0.06 | 0.82 $\pm$ 0.04 |
|          | Evasion   | 0.42 $\pm$ 0.08 | 0.36 $\pm$ 0.05 | 0.30 $\pm$ 0.05 |
| Citeseer | Clean     | ————            | 0.71 $\pm$ 0.04 | ————            |
|          | Poisoning | 0.69 $\pm$ 0.05 | 0.66 $\pm$ 0.03 | 0.62 $\pm$ 0.02 |
|          | Evasion   | 0.36 $\pm$ 0.06 | 0.30 $\pm$ 0.04 | 0.24 $\pm$ 0.05 |
| PubMed   | Clean     | ————            | 0.90 $\pm$ 0.04 | ————            |
|          | Poisoning | 0.88 $\pm$ 0.04 | 0.85 $\pm$ 0.05 | 0.81 $\pm$ 0.03 |
|          | Evasion   | 0.85 $\pm$ 0.07 | 0.82 $\pm$ 0.04 | 0.76 $\pm$ 0.03 |

attack.

We adopt the imperceptible attack strategies proposed by (Boucher et al. 2022) but with a key modification: we use a task-agnostic translation objective (English to French) to generate perturbations. The objective is to modify the input text such that the changes are visually identical but lead to an incorrect translation, thereby corrupting the encoded semantics. The intuition is that if the translation for a given text is wrong, the encoded semantics would also be wrong. We employ two imperceptible attacks: *Homoglyphs*, which replaces characters with visually similar ones and *Reorderings*, which use direction control characters to alter the rendering sequence. Examples of Homoglyphs and Reorderngs are in Appendix A.6.

Following (Boucher et al. 2022), we use differential evolution (Storn and Price 1997) to optimize these perturbations, maximizing the Levenshtein distance between the model’s output on the perturbed and unperturbed text. The perturbation budget is set to 10% of the average length of each textual feature.

**Results.** Table 4 shows that feature perturbations are dramatically more effective than the structural attacks studied above, and that Reordering is the more potent of the two feature attacks in every model $\times$ dataset $\times$ setting cell. Under evasion, Reordering slashes LLAGA’s accuracy by 73%, 72%, and 69% on Cora, Citeseer, and PubMed, roughly the same devastating range on all three datasets, and even under poisoning it removes 35%, 35%, and 26%, consistently exceeding the matched Homoglyph drops of 27%, 23%, and 20%. GRAPH-PROMPTER is again the more robust model in most cells (e.g., its Cora poisoning drops of 13% and 25% are well below LLAGA’s 27% and 35%), yet the key finding is that its earlier structural robustness does not carry over: Reordering evasion still erases 67%, 61%, and 53% of its accuracy. The overall impact is somewhat smaller on PubMed than on the two smaller datasets, echoing the density trend, but remains severe everywhere.

Placed side by side with the structural results in Figure 3,

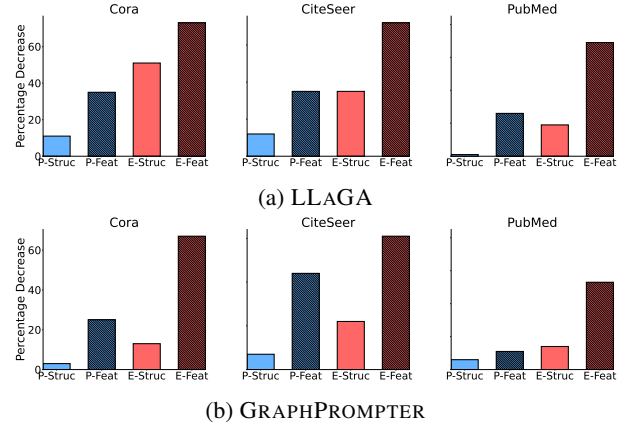


Figure 3: Structural vs. feature-based attacks on graph-aware LLMs, plotted as attack success, the percentage decrease in accuracy (y-axis). The feature bars (P-Feat, E-Feat) stand well above the matched structural bars (P-Struc, E-Struc) for both LLAGA and GRAPH-PROMPTER, visually confirming that feature perturbations are the more damaging channel. P-Struc, P-Feat, E-Struc, E-Feat denote poisoning/evasion attacks based on structural and feature perturbation, respectively.

these numbers mark a fundamental shift in where the vulnerability lies: feature-perturbation evasion (Reordering) outstrips structural evasion (METAATTACK) by a wide margin on *both* models, more than doubling the accuracy loss on GRAPH-PROMPTER in particular (from a 13% structural drop to a 67% feature drop on Cora). The consistency of this reversal across models and datasets indicates that graph-aware LLMs weight their textual input features far more heavily than graph structure, making the feature channel the dominant attack surface. We push this further by combining both channels in a unified attack in the next section.

## 8 A Unified Imperceptible Attack on Graph-aware LLMs

We propose a unified attack that combines both feature and structural perturbations to create a more imperceptible and effective adversarial strategy. We hypothesize that combining these attack types will more significantly degrade model accuracy than either method alone, while maintaining imperceptible alterations to the graph structure and features.

For the structural perturbation, we use METAATTACK, which modifies graph structure by subtly adjusting edges. For the feature component, we use a combination of Homoglyph and Reordering attacks, as described in previous section. Together, these perturbations create a highly imperceptible attack that simultaneously targets the relational and textual information on which graph-aware LLMs rely.

**Results.** Table 5 shows that jointly attacking both channels is the most damaging strategy in the entire study. Under evasion, the unified attack pushes LLAGA to 0.14, 0.10, and 0.12 (drops of 84%, 86%, and 87%) on Cora, Citeseer, and PubMed, and GRAPH-PROMPTER to 0.11, 0.13, and 0.19

Table 4: Accuracy under imperceptible feature-perturbation attacks (Homoglyph vs. Reordering), with the drop from clean accuracy in parentheses. Reordering is the stronger attack in every cell; evasion drops (up to 73%) far exceed poisoning drops, and both dwarf the structural-attack drops in Table 2.

| Data     | Method     | Poisoning             |                       | Evasion               |                       |
|----------|------------|-----------------------|-----------------------|-----------------------|-----------------------|
|          |            | LLAGA                 | GRAPHPROMPTER         | LLAGA                 | GRAPHPROMPTER         |
| Cora     | Homoglyph  | $0.65 \pm 0.05$ (27%) | $0.52 \pm 0.04$ (13%) | $0.41 \pm 0.04$ (54%) | $0.38 \pm 0.03$ (37%) |
|          | Reordering | $0.58 \pm 0.06$ (35%) | $0.45 \pm 0.05$ (25%) | $0.24 \pm 0.05$ (73%) | $0.20 \pm 0.03$ (67%) |
| Citeseer | Homoglyph  | $0.54 \pm 0.04$ (23%) | $0.65 \pm 0.04$ (7%)  | $0.32 \pm 0.03$ (55%) | $0.52 \pm 0.05$ (26%) |
|          | Reordering | $0.46 \pm 0.03$ (35%) | $0.39 \pm 0.02$ (44%) | $0.20 \pm 0.05$ (72%) | $0.27 \pm 0.02$ (61%) |
| PubMed   | Homoglyph  | $0.72 \pm 0.04$ (20%) | $0.83 \pm 0.04$ (8%)  | $0.36 \pm 0.06$ (60%) | $0.55 \pm 0.03$ (39%) |
|          | Reordering | $0.67 \pm 0.05$ (26%) | $0.80 \pm 0.03$ (11%) | $0.28 \pm 0.04$ (69%) | $0.42 \pm 0.04$ (53%) |

(82%, 81%, 79%), leaving both models effectively unusable. These figures exceed the best single-channel evasion attack, e.g., on Cora the 84% unified drop for LLAGA surpasses the 73% from Reordering alone and the 51% from METAATTACK alone, confirming that the two channels contribute complementary, not redundant, damage. Even the poisoning variant, at drops of 57–69%, is more severe than any single-channel poisoning result. The architectural takeaway is quantitatively clear: unlike traditional GNNs, which are chiefly sensitive to structural perturbations, graph-aware LLMs depend on the consistency of *both* structure and features, so simultaneously corrupting both, while remaining imperceptible, produces the largest measured degradation.

## 9 Ablation Study

### 9.1 Varying Feature Perturbation

To characterize how attack success scales with perturbation level, we sweep the fraction of perturbed features from 1% to 10%. The Reordering attack, which tracks the Homoglyph attack closely, exhibits a clean monotonic dose-response relationship on all models and datasets (Cora, Citeseer, and PubMed): fewer perturbed features yield a smaller accuracy drop, exactly as expected. The comparison across settings reveals a second, sharper pattern, at low perturbation levels the poisoning and evasion curves converge, whereas the large poisoning-vs-evasion gap we reported earlier only opens up at higher budgets. This is because a small adversarial signal is too weak to move the model regardless of whether it is injected at training (poisoning) or inference (evasion) time, so the two settings become indistinguishable in the low-budget regime. A detailed plot illustrating these trends is in the Appendix (Figure 2).

### 9.2 Does the Choice of LLM Affect Attack Performance?

To isolate the role of the backbone LLM, we hold the encoding architecture fixed and swap only the foundation model. LLAGA natively uses Vicuna-7B (Chiang et al. 2023) and GRAPHPROMPTER uses Llama2-7B (Touvron et al. 2023); we replace LLAGA’s backbone with Llama2-7B and re-run

Table 5: Accuracy under the unified imperceptible attack that combines structural and feature perturbations, with drop from clean accuracy in parentheses. Jointly attacking both channels is the most damaging setting overall, driving evasion accuracy as low as 0.10 (an 86% drop) and exceeding either channel alone.

| Data     | Attack    | LLAGA                 | GRAPHPROMPTER         |
|----------|-----------|-----------------------|-----------------------|
| Cora     | Poisoning | $0.35 \pm 0.05$ (61%) | $0.28 \pm 0.08$ (53%) |
|          | Evasion   | $0.14 \pm 0.06$ (84%) | $0.11 \pm 0.05$ (82%) |
| Citeseer | Poisoning | $0.26 \pm 0.04$ (63%) | $0.22 \pm 0.03$ (69%) |
|          | Evasion   | $0.10 \pm 0.09$ (86%) | $0.13 \pm 0.07$ (81%) |
| PubMed   | Poisoning | $0.39 \pm 0.07$ (57%) | $0.32 \pm 0.04$ (64%) |
|          | Evasion   | $0.12 \pm 0.05$ (87%) | $0.19 \pm 0.04$ (79%) |

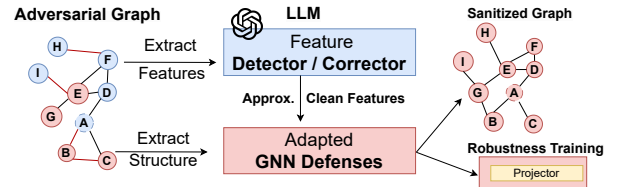


Figure 4: **Overview of GALGUARD defense.** Perturbed node features and edges are colored in blue and red, respectively. The LLM identifies perturbations in node features extracted from an adversarial graph and generates a corrected version. The refined features, together with the perturbed edges, are then passed through existing GNN defenses, which can either be used to obtain a sanitized graph or adapted for projectors’ robustness training.

all poisoning and evasion attacks on every dataset. The two backbones are statistically indistinguishable: clean accuracy differs by at most one point (Vicuna  $0.89 \pm 0.07$  vs. Llama2  $0.90 \pm 0.03$  on Cora), and the accuracy drops under attack track each other within one standard deviation across the board. The same equivalence holds when the backbone is swapped in GRAPHPROMPTER. This controlled comparison localizes the vulnerability firmly in the graph-encoding design rather than the choice of foundation LLM. The full results are in Table 2 of the Appendix.

## 10 Defense: GALGUARD

The measurements above expose a dual vulnerability, to both structural and feature perturbations, that existing GNN defenses cannot cover. Many such defenses rely on node feature similarity to prune or reweight edges (Wu et al. 2019; Dai et al. 2018; Zhang and Zitnik 2020; Jin et al. 2020; Zhang and Ma 2024), but they assume access to clean features, precisely the assumption our feature attacks violate; when features are perturbed these methods fail (see Figure 3 in the Appendix). Defenses that decouple structure and features (Shanthamallu, Thiagarajan, and Spanias 2021; Wu et al. 2021), and even OCR-based sanitization (Boucher et al. 2022), break down

under the coordinated attack that produced the 79–87% evasion drops reported earlier. Closing a gap of that magnitude requires a new, integrated defense.

To address this, we propose graph-aware LLM defense (GALGUARD), an end-to-end defense strategy combining two complementary approaches: (1) an LLM-based feature corrector and (2) adapted GNN-based structural defenses.

**Feature Robustness Strategy.** As shown in Figure 4, an LLM serves as an adaptive mechanism to analyze textual node features for anomalous patterns. It provides approximate corrections to restore feature integrity, thereby mitigating the effect of feature-level perturbations. This LLM-based correction is a practical solution for feature vulnerabilities. The specific instructions and prompts are in Appendix A.6. For our experiments, we employ GPT-4 Turbo as the LLM corrector.

**Graph Structure Robustness Strategy.** To enhance structural robustness, we employ a two-step approach. First, we adopt a *graph purification heuristic* (Wu et al. 2019) as a preprocessing step. For each node  $v$ , we compute the cosine similarity between its embedding  $\phi(x_v)$  and its neighbors’ embeddings  $\phi(x_u)$  and remove edges with low similarity. Second, we integrate *GNNGuard* (Zhang and Zitnik 2020). We directly apply GNNGuard’s principles to the message-passing layers of GRAPHPROMPTER. For LLAGA, we propose a novel adaptation: we incorporate the purification step into the construction of the computational tree for its node sequence. Furthermore, we introduce a learnable *global structural context embedding* ( $M_{global}$ ) as a learnable parameter in LLAGA’s projector. By concatenating  $M_{global}$  with the purified node sequence embedding and jointly training them, the model learns to operate robustly on potentially perturbed graph inputs. The plug-and-play design of GALGUARD’s allows other defenses, e.g. Pro-GNN (Jin et al. 2020), to be used instead of GNNGuard.

## 10.1 Results

Table 6 reports accuracy under METAATTACK alongside our two defense configurations: GALGUARD<sub>p</sub>, which pairs the LLM feature corrector with a graph purification module, and the full GALGUARD, which adds GNNGuard (or its LLAGA adaptation). The recovery is consistent across all datasets (Cora, Citeseer, PubMed), both models, and both attack types, and it is largest exactly where the attack was most damaging. The headline case is LLAGA evasion on Cora: METAATTACK had reduced accuracy to 0.44, GALGUARD<sub>p</sub> lifts it to 0.62, and the full GALGUARD restores it to 0.83, recovering nearly the entire gap back toward the 0.89 clean baseline. The two configurations form a clear ordering: GALGUARD<sub>p</sub> alone already provides a solid foundation (e.g., +0.18 on the Cora evasion case), while the full GALGUARD consistently adds a further margin on top (a further +0.21 in the same case), with every full-defense entry marked in bold as the best result in its cell. This monotonic METAATTACK < GALGUARD<sub>p</sub> < GALGUARD pattern quantifies the synergistic value of layering the GNN-inspired structural defense on top of feature correction.

Table 6: Accuracy under METAATTACK and the proposed defenses, for LLAGA and GRAPHPROMPTER under poisoning and evasion. GALGUARD<sub>p</sub> combines feature correction with graph purification; the full GALGUARD adds GNNGuard. Best result per column in **bold**. Recovery is consistent everywhere and follows METAATTACK < GALGUARD<sub>p</sub> < GALGUARD; the largest gain (Cora evasion, LLAGA: 0.44 → 0.83) occurs where the attack was strongest.

| Data     | Method                | LLAGA            |                  | GRAPHPROMPTER    |                  |
|----------|-----------------------|------------------|------------------|------------------|------------------|
|          |                       | Poisoning        | Evasion          | Poisoning        | Evasion          |
| Cora     | METAATTACK            | 0.79±0.03        | 0.44±0.06        | 0.58±0.05        | 0.52±0.04        |
|          | GALGUARD <sub>p</sub> | 0.82±0.05        | 0.62±0.04        | 0.59±0.03        | 0.56±0.04        |
|          | GALGUARD              | <b>0.85±0.04</b> | <b>0.83±0.04</b> | <b>0.60±0.05</b> | <b>0.59±0.03</b> |
| Citeseer | MetaAttack            | 0.63±0.04        | 0.50±0.03        | 0.65±0.06        | 0.56±0.03        |
|          | GALGUARD <sub>p</sub> | 0.64±0.03        | 0.56±0.02        | 0.67±0.05        | 0.60±0.06        |
|          | GALGUARD              | <b>0.67±0.04</b> | <b>0.62±0.04</b> | <b>0.68±0.04</b> | <b>0.64±0.05</b> |
| PubMed   | MetaAttack            | 0.89±0.08        | 0.73±0.04        | 0.85±0.02        | 0.77±0.03        |
|          | GALGUARD <sub>p</sub> | 0.90±0.06        | 0.78±0.03        | 0.85±0.04        | 0.81±0.04        |
|          | GALGUARD              | <b>0.90±0.01</b> | <b>0.87±0.02</b> | <b>0.89±0.02</b> | <b>0.89±0.01</b> |

## 11 Conclusion

In this paper, we take a substantial first step in quantifying the vulnerabilities of graph-aware LLMs to adversarial attacks, a largely unexplored area. Our controlled, head-to-head measurements pinpoint the graph-integration method as a critical source of vulnerability: the node sequence template in LLAGA is an exploitable attack surface, with our injection attacks driving evasion accuracy as low as 0.24 and beating standard structural attacks by up to 0.26 in absolute accuracy. The measurements further overturn the GNN-era assumption that structure is the main attack surface, feature-perturbation attacks remove up to 73% of accuracy on their own, far exceeding the up-to-51% structural drops, and a unified attack combining both channels pushes accuracy down by up to 87%, rendering graph-aware LLMs near-useless.

To address this, we propose an end-to-end defense, GALGUARD, which integrates an LLM-based feature corrector with adapted GNN-based structural defenses; it recovers accuracy consistently across all datasets and attack types, most strikingly restoring LLAGA’s Cora evasion accuracy from 0.44 to 0.83. As these models become more prevalent and hold potential in applications spanning social networks, healthcare, and finance, ensuring their resilience to adversarial exploitation is of the essence. We hope this work serves as a foundation for further research, inspiring deeper exploration into the vulnerabilities and defenses of graph-aware LLMs and fostering both theoretical insights and practical safeguards for their real-world deployment.

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## A Appendix

**Organization:** The Appendix is organized as follows. Appendix A.1 provides a detailed taxonomy of graph-aware LLMs. We discuss related work on attacks against Graph Neural Networks (GNNs) in Appendix A.2. Appendix A.3 gives an overview of the adversarial attacks, NETTACK and METAATTACK, while Appendix A.4 details the algorithm for the node sequence injection attack on LLAGA. We present the results of our ablation studies in Appendix A.5, including the effect of varying the amount of perturbed features and the choice of base LLM on attack performance. Finally, Appendix A.7 shows examples of feature attacks and the template for the LLM corrector, and Appendix A.8 discusses the implications of our research.

**Code.** Our code will be released upon acceptance.

### A.1 Graph Encoding-Based Taxonomy of LLM Adaptations

The integration of large language models (LLMs) with graph data has been an emerging area of research, with various approaches exploring how LLMs can be adapted or enhanced for graph-related tasks such as reasoning, representation, and inference. These approaches can be largely divided into two categories based on their methods for encoding graphs for LLMs: *textual descriptions of graphs*, where graph information is encoded as natural language, and *learned graph projector*, where a projector is utilized to encode graph data into embeddings or structured forms that LLMs can directly process. In this section, we describe the methods used in the different encoding approaches for integrating graph data into LLMs. Figure 5 provides the illustration of both approaches and Table 7 provides a summary of the differences between both approaches.

**Graph Representation as a Textual Description** In this first category, graph data is represented as natural language, allowing LLMs to process graph structures as textual descriptions. This approach has been explored extensively with works such as (Guo et al. 2023; Liu and Wu 2023; Ye et al. 2023; Wenkel, Wolf, and Knyazev 2023; Wei et al. 2024b; Fatemi, Halcrow, and Perozzi 2023; Tan et al. 2023; Zhao et al. 2023; Wei et al. 2024a) converting graph data into textual formats, often paired with specific instructions to query LLMs. For example, Tan et al. (2023) introduced a unified framework for attributed graph embedding by generating random walk-based textual sequences from graphs, which are then used to fine-tune LLMs. Similarly, Zhao et al. (2023) focused on graph reasoning within text spaces, translating graph structures into natural language to facilitate LLM interaction. Extending this idea, Wei et al. (2024a) augmented textual representations of graphs with visual representations of the graph, enhancing the reasoning capabilities of LLMs on graph data. In multimodal applications, Liu et al. (2023) and Su et al. (2022) linked molecule graphs with textual descriptions, enabling advances in drug discovery and molecular retrieval. Furthermore, Huang et al. (2023a) investigated how incorporating graph structure into LLM prompts can enhance predictive performance. Their findings revealed that LLMs tend to process graph-related prompts as contextual

paragraphs, rather than explicitly recognizing graph structures. This further underscores the importance of prompt design, as seen in studies like (Brannon et al. 2023; Zhang et al. 2023; Sun et al. 2023; Bi et al. 2024; Pan et al. 2024), which extended LLMs to heterogeneous and large-scale graph tasks. Complementing these efforts, (He et al. 2024) explored using LLM-generated explanations as features to enhance both the performance and interpretability of LLMs on graphs.

While still representing graphs as textual description, another line of work integrates pretraining strategies with GNNs to enhance the LLMs’ graph reasoning capabilities. For instance, Chen et al. (2023a) proposed two LLM-GNN integration approaches: 1) using LLMs to enhance node features for node classification, and 2) employing LLMs as direct predictors for graph tasks. In line with this, Chandra et al. (2020) utilized LLMs as text encoders and GNNs as graph structure encoders for tasks such as fake news detection in online communities. Similarly, Chen et al. (2023b) explored the possibility of performing node classification without explicit labels by using LLMs to annotate nodes, which are then used in training a GNN model. Likewise, Mavromatis et al. (2023) utilized LLM as a feature extractor for GNNs.

For inference, Zhu et al. (2024) proposed a fine-tuning method for textual graph that combined LLMs and GNNs through a tunable side structure alongside each layer of the LLM, enhancing both training efficiency and inference speed. Similarly, Duan et al. (2023) introduced SimTeG, a simple yet effective approach that enhances textual graph learning by leveraging LLMs. In their method, the LLM is first fine-tuned on a downstream task, such as node classification. Afterward, node embeddings are generated by extracting the last hidden states from the fine-tuned LLM. Additionally, Zhang (2023) proposed Graph-toolformer, which leverages prompt engineering, powered by ChatGPT, to augment LLMs’ graph reasoning capabilities.

While these approaches make LLMs more accessible for graph-related tasks, they often struggle to fully capture the structural intricacies inherent in graph data. Moreover, the performance of these methods remains heavily dependent on effective prompt engineering, which can be a challenging process, requiring careful design and experimentation to obtain optimal results.

**Graph Representation as a Learned Projector** To address the challenges posed by textual representation approaches, learned projector methods offer an alternative by transforming graph data into embeddings or graph vectors that LLMs can process more directly. For instance, Chen et al. (2024) introduced LLAGA, a large language and graph assistant that retains the general-purpose nature of LLMs while transforming graph structures into graph tokens, a format suitable for LLM processing. Similarly, (Liu et al. 2024; Qin et al. 2023) incorporated graph structure information into a pre-trained LLM through a tailored disentangled graph neural network layers. Tian et al. (2024) proposed using graph neural prompts to encode knowledge graph, transforming them into graph embeddings that LLMs can leverage during inference. Furthermore, Huang et al. (2023b) introduced a prompt-based node feature extractor (G-Prompt) for few-shot

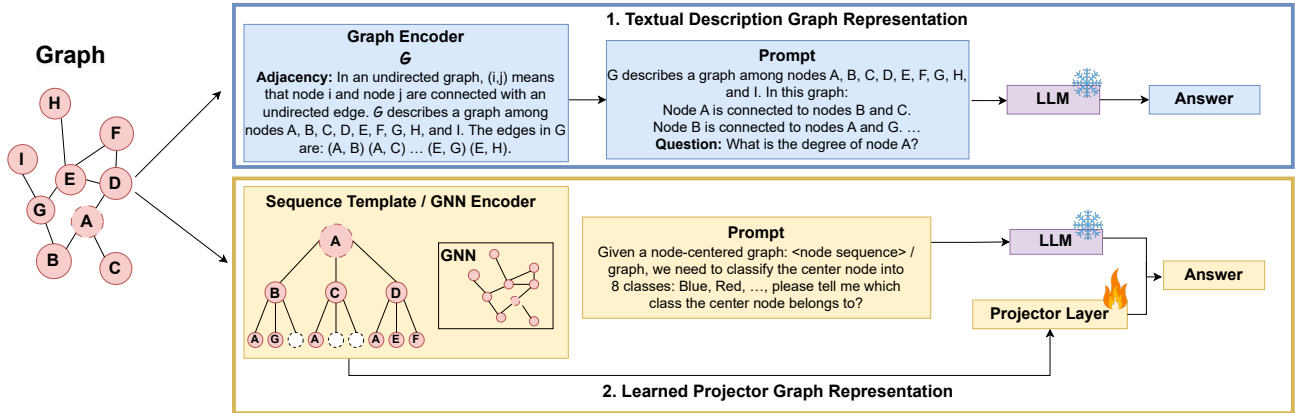


Figure 5: **Graph encoding-based adaptations of LLMs.** In the textual description graph representation, graphs are explained textually (top part). In the learned projector representation, graphs are encoded either by node templates that turns the graph into sequences or by GNNs (bottom part). Note that in both cases, the LLM is frozen. Only the projector that maps the graph into tokens is trained.

Table 7: Side-by-side comparison of the two graph encoding-based adaptations of LLMs across six design aspects, clarifying exactly where the textual-description and learned-projector approaches differ, and why we center our vulnerability study on the latter.

| Aspect                                | Textual Description Approach                                                | Learned Projector Approach                                                               |
|---------------------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| <b>Graph Encoding</b>                 | Converts graph data (nodes, edges) into natural language descriptions.      | Uses a learned projector to encode graph structures into embeddings or graph tokens.     |
| <b>LLM Modifications</b>              | No modification to the LLM.                                                 | No modification to the LLM.                                                              |
| <b>Graph Structure Representation</b> | Describes graph topology and relationships using textual narratives.        | Encodes relational and structural information directly into vector representations.      |
| <b>Flexibility</b>                    | Can only be used for simple graphs.                                         | More flexible in representing complex graph structures.                                  |
| <b>Computational Cost</b>             | No additional training is required. Lower cost.                             | Lightweight projector is trained, typically a linear layer. Minimal cost.                |
| <b>Suitability for Graph Tasks</b>    | May lose some relational nuance due to limitations in textual descriptions. | Captures detailed structural information, improving performance on graph-specific tasks. |

learning on text-attributed graphs. Their method combines a graph adapter (a learnable GNN layer) with task-specific prompts to capture neighborhood information and extract informative node features.

Another promising line of work condenses graph information into fixed-length prefixes using graph transformers, which are attached to each layer of the LLM to enhance its reasoning capabilities (Chai et al. 2023). Similarly, Yang et al. (2021) developed GraphFormers which seamlessly integrates text encoding and graph aggregation in an iterative workflow by nesting GNNs alongside the transformer blocks of a language model. Additionally, Peng et al. (2024) developed ChatGraph, an interactive interface that allows users to interact with their graphs through natural language. Here, graphs are represented as sequential paths and the LLM is fine-tuned on the graph data to support graph analysis.

In this paper, we do not consider projector adaptations that require retraining the LLM, as they significantly increase computational costs without yielding substantial performance gains. Instead, we focus on adaptation methods that utilize **frozen pre-trained LLMs** (Chen et al. 2024; Liu et al. 2024; Qin et al. 2023; Tian et al. 2024; Huang et al. 2023b). This approach not only reduces computational costs but also offers greater flexibility in integrating with various LLM models. Additionally, we can take advantage of LLMs’ extensive knowledge and capabilities without the overhead of retraining while still adapting them for specific graph tasks. Throughout this work, we employ two representative projector methods: **LLAGA** (Chen et al. 2024), which utilizes node-level templates and a linear projector, and **GRAPHPROMPTER** (Liu et al. 2024), which employs a GNN-based projector to encode the graph.

**This work.** In this work, we do not consider textual description methods, as they often struggle to fully capture the structural intricacies inherent in graph data. These methods are generally limited to simpler tasks, such as describing basic graph properties like node degree, and become infeasible for larger graphs due to scalability issues. In particular, the token limit in prompts restricts the detailed descriptions required for larger graph structures, quickly exceeding allowable token counts. Additionally, from a vulnerability perspective, textual description methods do not introduce any unique attack surface; since the graph is represented as simple text descriptions (e.g., “Node A is connected to Node B”), they are susceptible to traditional text-based adversarial attacks rather than the structural or feature perturbations specific to graph-based models. Therefore, they lack the complexity that would expose them to novel, graph-specific vulnerabilities, making them less relevant for our investigation into adversarial robustness of graph-aware LLMs.

## A.2 Attacks on GNN

Several types of attacks exist on GNNs, including membership inference attacks (Olatunji, Nejdli, and Khosla 2021; He et al. 2021; Conti et al. 2022), which aim to determine whether a specific node was part of the model’s training set; attribute inference attacks (Gong and Liu 2018; Olatunji et al. 2023a; Duddu, Boutet, and Shejwalkar 2020), which exploit the GNN’s output and learned embeddings to infer

sensitive or missing node attributes; property inference attacks (Zhang et al. 2022; Wang and Wang 2022), which seek to extract global properties of the entire graph; adversarial attacks (Zügner, Akbarnejad, and Günnemann 2018; Zügner and Günnemann 2019; Li et al. 2023; Dai et al. 2018; Sun et al. 2022), which attempt to degrade the model’s performance by introducing subtle perturbations to the graph or node features; and graph reconstruction attacks (Zhang et al. 2021; Olatunji et al. 2023b), which aim to reconstruct the underlying graph structure using the learned embeddings or model outputs.

Among these attack types, adversarial attacks are of particular concern due to their direct impact on the model’s performance. In adversarial attacks, small, carefully crafted *perturbations* are introduced into the graph, such as modifying node features or altering edges, with the goal of significantly degrading the GNN’s performance. These attacks can be categorized into two types: **poisoning attacks**, which occur during the *training phase* by manipulating the training data, and **evasion attacks**, which occur at *test time*, where the adversary introduces perturbations to the input graph to mislead the model’s predictions. For instance, Zügner, Akbarnejad, and Günnemann (2018) demonstrated that adversarial perturbations on graph structures or node features can significantly degrade the performance of GNNs. Their method, **NETTACK**, employed a greedy optimization technique to compute the minimal perturbations that would maximize the model’s loss function during training. This approach effectively identifies the smallest changes in the features or graph structure needed to disrupt the model’s predictions. Li et al. (2023) revisited the problem of adversarial attacks on graph data from a data distribution perspective, highlighting how adversarial examples can significantly alter the underlying data distribution and proposed a novel defense mechanism to mitigate such effects. Similarly, Wu et al. (2019) provided a comprehensive analysis of adversarial examples on graph data, offering deep insights into both attacks and defenses. They focused on the vulnerabilities of GNNs to small perturbations and proposed defense techniques designed to detect and counteract adversarial modifications, thus enhancing model robustness. **METAATTACK** (Zügner and Günnemann 2019) performed both poison and evasion attacks on GNNs by treating the graph structure as a hyperparameter that can be optimized. The attack directly modifies the graph structure using gradient descent, formulating the problem as a bi-level optimization task. Their method utilized meta-gradients, which capture how small perturbations in the graph structure affect the attacker’s loss after training, enabling precise updates to the graph that degrade the model’s performance.

In this paper, we leverage **NETTACK** (Zügner, Akbarnejad, and Günnemann 2018) and **METAATTACK** (Zügner and Günnemann 2019) as representative adversarial attack methods, encompassing both poisoning and evasion strategies, to conduct our investigation. Specifically, poisoning attacks target model integrity during training, while evasion attacks compromise performance at inference or test time.

### A.3 How NETTACK and METAATTACK Work

NETTACK performs graph structure perturbation by adding and removing edges between nodes, while ensuring that the degree of the perturbed graph is preserved. It uses a greedy optimization technique to compute the minimal perturbations that will maximize the model’s loss function during training, with respect to a surrogate model. This surrogate model is a linearized 2-layer graph convolutional network (GCN), which approximates how changes in the graph structure affect the classification result. The primary objective of NETTACK is to increase the model’s confidence in incorrect classifications by maximizing the distance between the logits of the perturbed graph and those of the clean graph. To achieve this, NETTACK identifies candidate edges for perturbation around a target node and evaluates a score function, for each connected edge. This score indicates how significantly altering a particular edge will impact the classification result of the target node. The edge with the highest score is then selected, and the graph is updated by either adding or removing that edge. Since NETTACK is a targeted attack (only cause misclassification of a specific single node), we adapt the same method in (Zügner and Günnemann 2019) to convert it into untargeted attack (compromise node classification performance of the model) which randomly select one test node and attack it using NETTACK while considering all nodes in the network. We then aggregate the result over the test nodes.

METAATTACK optimizes the graph structure directly via gradient descent by treating the adjacency matrix (graph structure) as a hyperparameter. It frames the attack as a bi-level optimization problem: the inner level minimizes the training loss on the original graph, while the outer level minimizes the loss on the poisoned graph. To achieve this, METAATTACK utilizes a surrogate model, specifically a linearized two-layer GCN, to approximate how changes in the graph structure affect model performance. A score function evaluates the impact of each perturbation, and the attack flips the sign of meta-gradients (gradient wrt hyperparameters) for connected nodes, indicating which edges to remove. The algorithm then greedily selects perturbations—either adding or removing edge based on those with the highest scores, to maximize the attack’s impact.

For both methods, we use the same unnoticeability constraint introduced by (Zügner, Akbarnejad, and Günnemann 2018), which ensures that the graph’s degree distribution only changes slightly. Therefore, we only perturb 10% of the edges.

In our poisoning attack experiments with LLAGA, we employ the perturbed graph structure obtained from NETTACK or METAATTACK to generate the node-level sequences and retrain the projector that converts the node-level sequences into a sequence of token embeddings. Similarly, for GRAPH-PROMPTER, we utilize the perturbed graph structure, re-encode the graph using a graph attention network (GAT), and retrain the projector with the perturbed graph. For evasion attacks, the graph-aware LLM is trained on unperturbed data, with perturbations introduced only during the inference stage.

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Algorithm 1: Injection Attacks on Neighborhood Detail Template for Obtaining Node Sequence.

---

**Input:** Graph  $G = (V, E)$ , target node  $u$ , tree depth  $d$ , children size  $k$

**Output:** Modified graph  $G'$  with injected nodes

```

1: procedure NEIGHBORHOODDETAILTEMPLATE( $u, d, k$ )
2:    $T \leftarrow \emptyset$  ▷ Initialize empty tree
3:    $T.\text{root} \leftarrow u$ 
4:   for level  $\ell = 1$  to  $d$  do
5:     for each node  $v$  at level  $\ell - 1$  do
6:        $N_v \leftarrow \text{Sample}(\mathcal{N}(v), k)$  ▷ Sample  $k$ 
       neighbors
7:       if  $|N_v| < k$  then
8:         Fill remaining slots with placeholders
9:       end if
10:      Add  $N_v$  as children of  $v$  in  $T$ 
11:    end for
12:  end for
13:  return  $T$ 
14: end procedure

15: procedure ATTACK( $G, u, d, k, \text{AttackStrategy}$ )
16:   $G' \leftarrow G$  ▷ Create copy of original graph
17:   $T \leftarrow \text{NEIGHBORHOODDETAILTEMPLATE}(u, d, k)$ 
18:   $P \leftarrow \text{GetPlaceholders}(T)$ 
19:  if  $|P| > 0$  then
20:     $\mathcal{N}_2 \leftarrow \text{2-hop neighborhood of } u$ 
21:    if  $\text{AttackStrategy} = \text{NI}$  then
22:       $V_{\text{non-adjacent}} \leftarrow V \setminus \mathcal{N}_2$ 
23:       $V_{\text{inject}} \leftarrow \text{Sample}(V_{\text{non-adjacent}}, |P|)$ 
24:      for each placeholder position  $p \in P$  do
25:         $v \leftarrow \text{Next node from } V_{\text{inject}}$ 
26:        Add edge  $(\text{parent}(p), v)$  to  $G'$ 
27:      end for
28:    else if  $\text{AttackStrategy} = \text{SI}$  then
29:       $V_{\text{non-adjacent}} \leftarrow V \setminus \mathcal{N}_2$ 
30:       $s \leftarrow \text{GetHighestDegreeNode}(V_{\text{non-adjacent}})$ 
31:      for first placeholder position  $p \in P$  do
32:        Add edge  $(\text{parent}(p), s)$  to  $G'$ 
33:        break ▷ Only use first placeholder
34:      end for
35:    else if  $\text{AttackStrategy} = \text{MSI}$  then
36:       $V_{\text{non-adjacent}} \leftarrow V \setminus \mathcal{N}_2$ 
37:       $S \leftarrow \text{GetNodesOrderedByDegree}(V_{\text{non-adjacent}})$ 
38:      for  $i = 1$  to  $|P|$  do
39:         $s \leftarrow \text{Get } i\text{-th node from } S$ 
40:         $p \leftarrow \text{Get } i\text{-th placeholder from } P$ 
41:        Add edge  $(\text{parent}(p), s)$  to  $G'$ 
42:      end for
43:    end if
44:  end if
45:  return  $G'$ 
46: end procedure

```

---

### A.4 Algorithm for Node Sequence Template Injection Attacks

We present the algorithm for the node sequence template injection attack in LLAGA in Algorithm 1. The model con-

structs a *fixed-shape* computational tree for each node  $u$ , sampling  $k$  neighbors per node up to depth  $d$  (lines 1–14). When a node has fewer than  $k$  neighbors, placeholders are inserted, creating an attack surface. We exploit this by injecting malicious nodes into placeholder positions through graph manipulation. We propose three strategies: *Non-Adjacent Injection (NI)* inserts nodes outside the 2-hop neighborhood of  $u$  (lines 21–27); *Supernode Injection (SI)* injects a high-degree node into one placeholder (lines 29–33); and *Multiple Supernode Injection (MSI)* distributes distinct high-degree nodes across all placeholders (lines 35–43). These uninformative nodes corrupt the node sequence, degrading the projection layer’s performance. The inclusion of uninformative nodes in the sequence corrupts the target’s representation, impairing training of the projection layer.

## A.5 Ablation Studies

Here, we present the results for the ablation studies presented in the main paper.

**Ablation Results: Varying Feature Perturbation** Figure 6 plots the accuracy of LLGA and GRAPH-PROMPTER as the fraction of perturbed features increases. For both models the trend is monotonic and consistent: more perturbed features means lower accuracy, with no plateau or reversal, quantifying a smooth dose-response sensitivity to feature-level attacks across the full 1%–10% range.

**Ablation Results: Does the Choice of LLM Affect Attack Performance?** Table 8 isolates the effect of the base LLM within the LLGA framework, and the two backbones are quantitatively interchangeable on every axis. Across Cora, Citeseer, and PubMed, Vicuna-7B and Llama2-7B match to within about one point in clean accuracy and show the same degradation profile under both poisoning and evasion. The tightest illustration is METAATTACK evasion on Cora,  $0.44 \pm 0.06$  (Vicuna-7B) vs.  $0.43 \pm 0.08$  (Llama2-7B), a 0.01 difference that sits well inside one standard deviation. Every remaining dataset×attack cell shows a similarly negligible gap. The comparison therefore attributes attack effectiveness to the encoding design, not the base LLM.

## A.6 Results on the ArXiv Dataset

We now scale the evaluation to the large ArXiv dataset (169,343 nodes, 1,166,243 edges, sparsity 0.81) to test the node sequence template attacks on a large, real-world graph. Table 9 reveals a striking reversal of the poisoning-vs-evasion ordering seen on the smaller datasets: on ArXiv, poisoning is the *stronger* setting across all three injection strategies, cutting accuracy to 0.50/0.43/0.42 (NI/SI/MSI) versus 0.71/0.69/0.60 under evasion, whereas on Cora, Citeseer, and PubMed evasion was consistently the more damaging. This inversion indicates that ArXiv’s structure makes it more vulnerable to poisoning yet comparatively more robust to evasion. Despite its very low density, a single injected supernode is still enough to move accuracy substantially (an 0.04 absolute drop from SI to MSI even under evasion), showing the template attack remains effective at scale.

## A.7 Feature Attack Examples and Defense Template

**Feature Attack Examples** The examples in the prompt, as shown in Appendix A.7, illustrate our adversarial attacks on real features from the Cora dataset. The “input” is the text received by the LLM where the attack is being applied. It is important to note that while a human reader perceives the original text and the perturbed text as visually identical or very similar, the LLM processes a sequence of different Unicode characters or a different character order, which can alter its behavior. We show this in Table 10 for homoglyph and Table 11 for reordering attack.

**Feature Corrector Defense Template** We designed a feature corrector module to mitigate the effects of imperceptible attacks, specifically homoglyph substitutions and character reorderings. This module is implemented as a prompt-based defense using LLMs. The prompt is carefully crafted to instruct the LLM to act as a linguistic validator, focusing on detecting and correcting subtle, human-imperceptible perturbations.

The template for our prompt is shown below. It explicitly states the task and provides a clear output format, which significantly improves the LLM’s performance.

## A.8 Detailed Discussion

The findings of our work raise several critical insights into the vulnerabilities and design considerations of graph-aware LLMs. As the integration of graph-structured data into LLMs continues to gain traction for graph tasks, it is important to weigh not only the reported performance benefits but also the measured robustness costs of these novel architectures. Our systematic, quantitative comparison of adversarial attacks on two representative graph-aware LLMs, LLGA and GRAPH-PROMPTER, exposes large and consistent weaknesses under the poisoning and evasion attacks traditionally used against GNNs, with accuracy drops reaching 51% for structural attacks and up to 87% once feature perturbations are included. Our key observations are discussed below.

**Impact of Graph Encoding on Adversarial Vulnerability** The key observation from our results is that the choice of graph-integration method measurably changes a model’s susceptibility to adversarial attacks. The node sequence template in LLGA is the clearest example: it opens an attack surface, unique to LLGA, that our injection attacks exploit to push evasion accuracy as low as 0.24, a larger drop than any standard structural attack achieves on the same model. This crystallizes a critical design tradeoff, node-level templates incorporate graph structure effectively and yield strong clean-task performance, but they simultaneously introduce attack vectors absent in traditional GNNs. By contrast, GRAPH-PROMPTER’s GNN-based projector is consistently the more resilient encoder under structural attack (e.g., 13% vs. LLGA’s 51% drop under METAATTACK evasion on Cora), quantifying the robustness advantage of hybrid architectures that combine the strengths of LLMs and GNNs.

**Sensitivity of Graph-aware LLMs to Feature Perturbations** Beyond the encoding-specific weakness, our

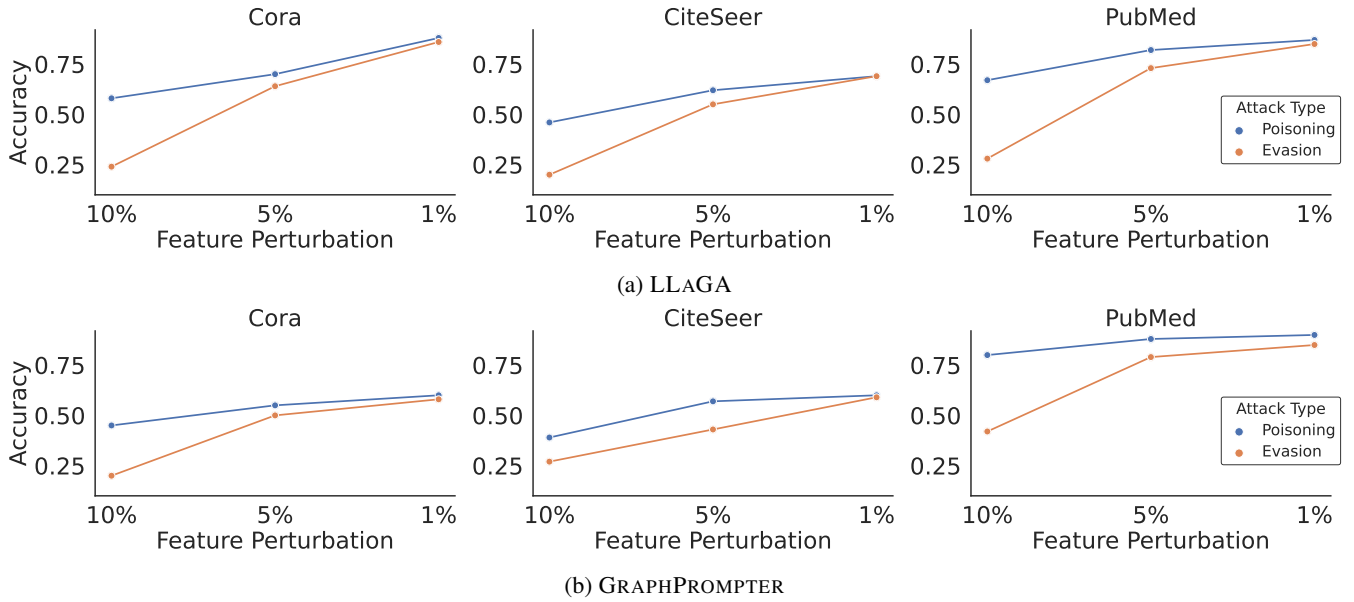


Figure 6: Accuracy of the graph-aware LLMs as the percentage of perturbed features is varied. Accuracy falls monotonically with the perturbation fraction for both models, so from the attacker’s perspective, lower accuracy indicates a stronger attack.

Table 8: Effect of swapping LLaGA’s base LLM (Vicuna-7B vs. Llama2-7B) on clean accuracy and on accuracy under each attack. Paired rows match within roughly one point, and all differences fall inside one standard deviation, showing that susceptibility is essentially invariant to the choice of base model.

| Data     | Base Model | Clean           | Poisoning       |                 |                 | Evasion         |                 |                 |
|----------|------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
|          |            |                 | NETTACK         | METAATTACK      | MSI             | NETTACK         | METAATTACK      | MSI             |
| Cora     | Vicuna-7B  | $0.89 \pm 0.07$ | $0.87 \pm 0.04$ | $0.79 \pm 0.03$ | $0.82 \pm 0.04$ | $0.55 \pm 0.09$ | $0.44 \pm 0.06$ | $0.30 \pm 0.05$ |
|          | Llama2-7B  | $0.90 \pm 0.03$ | $0.86 \pm 0.02$ | $0.77 \pm 0.05$ | $0.82 \pm 0.02$ | $0.56 \pm 0.07$ | $0.43 \pm 0.08$ | $0.31 \pm 0.01$ |
| Citeseer | Vicuna-7B  | $0.71 \pm 0.04$ | $0.64 \pm 0.05$ | $0.63 \pm 0.04$ | $0.62 \pm 0.02$ | $0.59 \pm 0.04$ | $0.50 \pm 0.03$ | $0.24 \pm 0.05$ |
|          | Llama2-7B  | $0.71 \pm 0.05$ | $0.65 \pm 0.03$ | $0.60 \pm 0.03$ | $0.61 \pm 0.04$ | $0.58 \pm 0.03$ | $0.48 \pm 0.05$ | $0.26 \pm 0.03$ |
| PubMed   | Vicuna-7B  | $0.90 \pm 0.04$ | $0.89 \pm 0.03$ | $0.89 \pm 0.08$ | $0.81 \pm 0.03$ | $0.84 \pm 0.05$ | $0.73 \pm 0.04$ | $0.76 \pm 0.03$ |
|          | Llama2-7B  | $0.91 \pm 0.06$ | $0.88 \pm 0.04$ | $0.90 \pm 0.02$ | $0.80 \pm 0.04$ | $0.85 \pm 0.03$ | $0.75 \pm 0.02$ | $0.77 \pm 0.02$ |

strongest quantitative signal is the effectiveness of feature-perturbation attacks against *both* graph-aware LLMs. Regardless of how each model encodes structure, imperceptible node-feature perturbations remove up to 73% of accuracy under evasion, exceeding the up-to-51% structural drops and doing so on both LLaGA and GRAPHPROMPTER. This points to a broader property of the LLM paradigm: these models excel at handling textual data, but that very reliance on textual features makes feature-directed adversarial perturbations, rather than structural ones, the dominant and still-open robustness challenge. The ability of adversaries to degrade model performance by subtly altering node attributes without being detected calls into question the integrity of graph-aware LLMs, especially in security-sensitive applications. While text sanitization techniques, such as those discussed by (Boucher et al. 2022), offer potential defenses, addressing these imperceptible attacks such as homophony-based manipulations remains a complex challenge. As proposed, using LLMs as a corrector is one way to mitigate such vulnerabili-

ties, by prompting the model to detect and repair suspicious or malformed inputs before downstream processing.

**Data-dependent Variability of Poisoning and Evasion Attacks on Graph-aware LLMs** Our measurements show that attack efficacy is not uniform across datasets but tracks graph characteristics in a predictable way. On Cora, CiteSeer, and PubMed, evasion is consistently the more damaging setting and denser graphs degrade more, yet on the large ArXiv graph this ordering inverts and poisoning becomes the stronger attack. This dataset-dependent swing underscores that any single vulnerability figure is incomplete without the accompanying graph statistics, and that robustness evaluations must span a range of densities and scales. As these models are increasingly deployed in diverse real-world settings, their vulnerabilities must be assessed not only in terms of algorithmic design but also in relation to the specific domains and data they are applied to.

### Prompt for Feature Corrector

You are a linguistic validator. Your task is to detect and correct any subtle, imperceptible text perturbations such as homoglyph substitutions (e.g., replacing "o" with "0") and character reorderings. These changes are often used to attack machine learning models. Given a text string, carefully analyze it. If you identify any perturbations, provide the corrected, original-looking version of the text. If no perturbations are found, return the original text unchanged.

#### Example 1 (Homoglyph Substitution):

**Input:** Title: The mgprir heuristic fr discvring prtin squnc pttrens Abstract: Svrl cmputr lgrithms fr discvring pttrens in grups f prtin squncs r in us tht r bsd on fitting th prmtrs f sttisticl mdl to grup of rlted squncs.

**Output:** Title: The megaprior heuristic for discovering protein sequence patterns Abstract: Several computer algorithms for discovering patterns in groups of protein sequences are in use that are based on fitting the parameters of a statistical model to a group of related sequences.

#### Example 2 (Character Reordering):

**Input:** Title: Appliatcions of machine lreanign: a medical foall-up study Abstract: This ppaer dscribes pmerlinary work that aims to aplpy smoe learning sttraegies to a meidcal flolow-up study. An inestivagiottn of the apclpiation of three machine larening algorithms-lR, FOIL and InductH to identify risk frcatos that govern the clopsuspension cure rtae has been mdae.

**Output:** Title: Applications of machine learning: a medical follow up study Abstract: This paper describes preliminary work that aims to apply some learning strategies to a medical follow-up study. An investigation of the application of three machine learning algorithms-lR, FOIL and InductH to identify risk factors that govern the colposuspension cure rate has been made.

#### Example 3 (Mixed Homoglyph & Reordering):

**Input:** Title: Submitted to NIPS96, Sctin: Applications. Prfrnc: Oral prsentatin Reinforcement Lrning for Dynmic Channl Allocation in Abstract: In ellulr tlphne systms, n imrtnt prblm is to dynmiclly allcoct th cmmunictin rsurc (chnnls) so as to mximize srvc in stchstic cllr nvirnmt. This prblem is naturally frmultd s dynmic prgrmming prblm and we us a reinforcement lerning (RL) mthd to find dynmic chnnl llction plicies that ar bttr than prvius heuristic slutins.

**Output:** Title: Submitted to NIPS96, Section: Applications. Preference: Oral presentation Reinforcement Learning for Dynamic Channel Allocation in Abstract: In cellular telephone systems, an important problem is to dynamically allocate the communication resource (channels) so as to maximize service in a stochastic caller environment. This problem is naturally formulated as a dynamic programming problem and we use a reinforcement learning (RL) method to find dynamic channel allocation policies that are better than previous heuristic solutions.

**Your Task:** Text to correct: {perturbed\_text\_here}

Corrected Text:

Figure 7: The prompt template used for the LLM-based feature corrector module. It explicitly defines the role, task, and expected output format for the LLM.

Table 9: Node sequence template injection attack including the large-scale ArXiv dataset (accuracy). *NI*, *SI*, *MSI* denote non-adjacent, single-supernode, and multiple-supernode injection; Clean is the un-attacked baseline. ArXiv reverses the trend of the smaller datasets, poisoning is more damaging than evasion there, while the  $NI < SI < MSI$  severity ordering still holds throughout.

| Data     | Attack    | NI              | SI              | MSI             |
|----------|-----------|-----------------|-----------------|-----------------|
| Cora     | Clean     | ——              | $0.89 \pm 0.07$ | ——              |
|          | Poisoning | $0.86 \pm 0.03$ | $0.84 \pm 0.06$ | $0.82 \pm 0.04$ |
|          | Evasion   | $0.42 \pm 0.08$ | $0.36 \pm 0.05$ | $0.30 \pm 0.05$ |
| Citeseer | Clean     | ——              | $0.71 \pm 0.04$ | ——              |
|          | Poisoning | $0.69 \pm 0.05$ | $0.66 \pm 0.03$ | $0.62 \pm 0.02$ |
|          | Evasion   | $0.36 \pm 0.06$ | $0.30 \pm 0.04$ | $0.24 \pm 0.05$ |
| PubMed   | Clean     | ——              | $0.90 \pm 0.04$ | ——              |
|          | Poisoning | $0.88 \pm 0.04$ | $0.85 \pm 0.05$ | $0.81 \pm 0.03$ |
|          | Evasion   | $0.85 \pm 0.07$ | $0.82 \pm 0.04$ | $0.76 \pm 0.03$ |
| ArXiv    | Clean     | ——              | $0.73 \pm 0.02$ | ——              |
|          | Poisoning | $0.50 \pm 0.03$ | $0.43 \pm 0.04$ | $0.42 \pm 0.03$ |
|          | Evasion   | $0.71 \pm 0.06$ | $0.69 \pm 0.02$ | $0.60 \pm 0.03$ |

**Implications for Graph-aware LLM Development** The vulnerabilities quantified in this study have broader implications for the future development and deployment of graph-aware LLMs. Given their increasing adoption for tasks that were traditionally the domain of GNNs, and given that the same models that report accuracy gains also incur the largest measured robustness losses, it is crucial to weigh this performance-versus-robustness tradeoff explicitly when designing these paradigms. As graph-aware LLMs continue to evolve, there is a pressing need for the development of more robust encoding techniques and defenses against adversarial manipulation, particularly in the context of feature perturbations. Furthermore, understanding how adversarial attacks transfer from GNNs to graph-aware LLMs opens new avenues for research in protecting these models against sophisticated adversaries.

**Difficulty in Applying Existing GNN defenses Against Adversarial Attacks on Graph-aware LLMs** **Defenses on adversarial attacks based on feature perturbation.** Adversarial attacks on GNNs have predominantly focused on structural perturbations, as feature-based perturbations typically fail to influence the predicted class of the target node (Dai et al. 2018; Wu et al. 2019; Zügner and Günnemann 2019), making them ineffective attack strategies. Consequently, there are *no known defenses against feature-based attacks on GNNs*, given their limited effectiveness. The graph-aware LLM paradigm overturns this premise: as our main-paper measurements show, feature-perturbation attacks are not merely effective here but the single most damaging class, removing up to 73% of accuracy on their own. This introduces a novel challenge within the graph-aware LLM framework, where feature-based attacks, previously ineffective against GNNs, can now successfully manipulate model behavior. This shift highlights the need

for developing new defenses that address both structural and feature-based adversarial perturbations. While text sanitization techniques, such as the use of optical character recognition (OCR) as a pre-processing step for encoding text for the LLM, offer potential defenses, addressing these imperceptible attacks such as homoglyph-based manipulations remains a complex challenge as OCR is imperfect and tends to misinterpret homoglyph at a higher rate than unperturbed text (Boucher et al. 2022). This led to our design of self-correcting mechanism for feature perturbation using LLMs.

**Defenses on adversarial attacks based on structural perturbation.** Due to the transferability of structure-based adversarial attacks from GNNs to graph-aware LLMs, particularly in evasion attacks, it is natural to expect that defenses developed for GNNs would also be effective in this paradigm. However, this assumption does not hold. Unlike traditional GNN settings, *graph-aware LLMs are also susceptible to feature perturbations, rendering these defenses ineffective.*

One such defense mechanism in GNNs involves making the adjacency matrix trainable, allowing edge weights to adapt during training to mitigate adversarial perturbations. This approach assigns lower weights to edges connecting dissimilar nodes and higher weights to edges linking nodes with greater feature similarity. Several **preprocessing-based** defenses have been proposed building on this approach, such as removing edges between nodes with low feature similarity (Wu et al. 2019) or introducing randomized edge dropping (Dai et al. 2018). However, these defenses rely heavily on clean, unperturbed node features to compute reliable similarity scores. When node features are also perturbed, as in our attacks, such defenses not only become ineffective but can actively worsen the problem by wiring up incorrect edges. Figure 8 makes the failure concrete on a random Cora node: reconstruction from clean features preserves at least some original edges (while adding a few spurious ones), whereas reconstruction from perturbed features preserves *none* of the original connections, a complete collapse of the similarity signal these defenses depend on.

Another line of work focuses on decoupling reliance on the graph structure. For instance, (Shanthamallu, Thiagarajan, and Spanias 2021) proposed a defense that trains a surrogate predictor dependent solely on node features and employs an uncertainty matching strategy to extract graph information from the GNN. Similarly, Wu et al. (2021) introduced a co-training framework that trains submodels independently on feature and structural views, allowing knowledge distillation through the inclusion of confident unlabeled data into the training set. However, these approaches also assume access to unperturbed node features, an assumption that does not hold in our setting, where both structure and features are adversarially perturbed. Consequently, structural defense strategies developed for GNNs are not directly applicable to the graph-aware LLM paradigm, which calls for novel defenses tailored to this paradigm.

Table 10: An illustration of the imperceptible homoglyph attack. While the perturbed text is processed as distinct characters by the LLM, the visual output is nearly identical for a human, as shown in the image.

| Original Feature                                                                                                                                                                                                                                                                                 | What LLM Sees                                                                                                                                                                                                                                                       | What a Human Sees                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Title: The megaprior heuristic for discovering protein sequence patterns</p> <p>Abstract: Several computer algorithms for discovering patterns in groups of protein sequences are in use that are based on fitting the parameters of a statistical model to a group of related sequences.</p> | <p>Title: The mgprir<br/>huristic fr discvring<br/>prtln squnc pttrns</p> <p>Abstract: Svrl cmputr<br/>lgrithms fr discvring<br/>pttrns in grups f prtln<br/>squncs r in us tht r bsd<br/>on fitting th prmtrs f<br/>stttisticl mdl to grup of<br/>rltd sauncs.</p> | <p>Title: The megaprior heuristic for discovering protein sequence patterns.</p> <p>Abstract: Several computer algorithms for discovering patterns in groups of protein sequences are in use that are based on fitting the parameters of a statistical model to a group of related sequences.</p> |

Table 11: An illustration of the imperceptible reordering attack. While the perturbed text is processed as distinct characters by the LLM, the visual output is nearly identical for a human, as shown in the image.

| Original Feature                                                                                                                                                                                                                                                                                                                                                                            | What LLM Sees                                                                                                                                                                                                                                                                                                                                                                                                                                   | What a Human Sees                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Title: Applications of machine learning: a medical follow up study.</p> <p>Abstract: This paper describes preliminary work that aims to apply some learning strategies to a medical follow-up study. An investigation of the application of three machine learning algorithms-1R, FOIL and InductH to identify risk factors that govern the colposuspension cure rate has been made.</p> | <p>Title: Appliatcions of<br/>machine lreanign: a<br/>medical foall-up study</p> <p>Abstract: This ppaer<br/>dsceribes pmerlinary<br/>work that aims to aplpy<br/>smoe learning sttraegies<br/>to a meidcal flolow-up<br/>study. An inestivagiotn<br/>of the aplcpiation of<br/>three machine larening<br/>algorithms-1R, FOIL and<br/>InductH to identify risk<br/>frcatos that govern the<br/>clopsuspension cure rtae<br/>has been mdae.</p> | <p>Title: Applications of machine learning: a medical follow up study</p> <p>Abstract: This paper describes preliminary work that aims to apply some learning strategies to a medical follow-up study. An investigation of the application of three machine learning algorithms-1R, FOIL and InductH to identify risk factors that govern the colposuspension cure rate has been made.</p> |

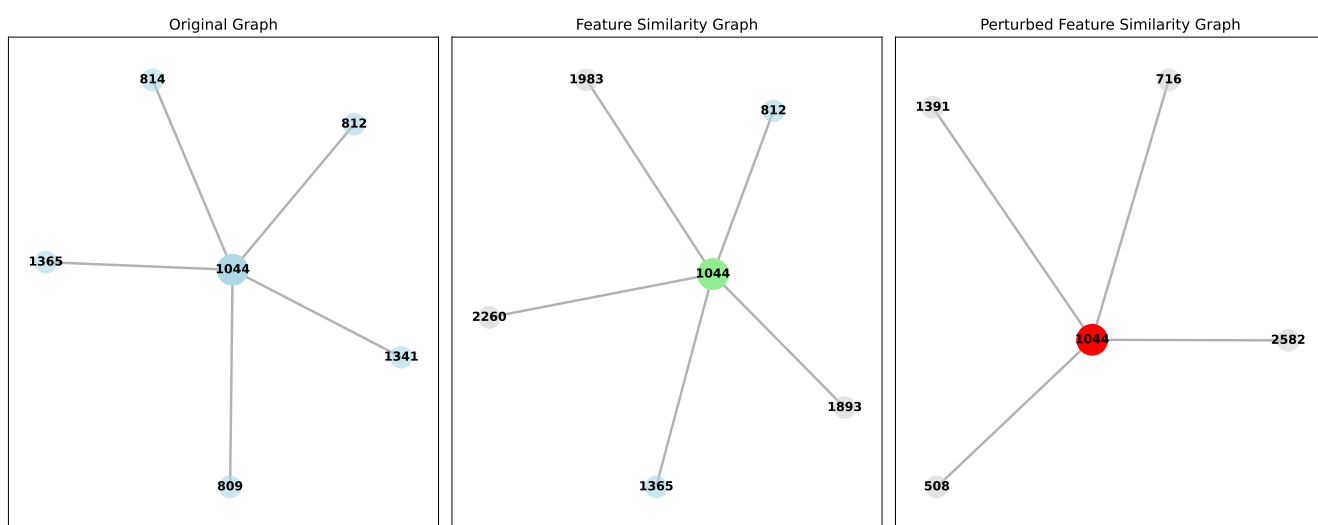


Figure 8: Edges reconstructed for a random node 1044, illustrating why similarity-based defenses collapse under feature perturbation. **Original (left)**: ground-truth edges. **Feature Similarity (middle)**: reconstruction from clean features, with blue nodes marking connections correctly preserved from the original graph. **Perturbed Feature Similarity (right)**: reconstruction from perturbed features, which preserves none of the original edges.